Definition: A network is an important variant of a graph, where edges are:

* Directed
* Weighted



Examples:
* liquid flow
* computer networks

More generally: entities and relationship.

A digraph (directed graph) D, consists of a graph $G = (V, E)$ in which edges (arcs) have a direction, so that each (u, v) ($u \to v$) has an initial vertex u, and a terminal vertex v.

Note: Edges (u, v) and (v, u) are two different edges.

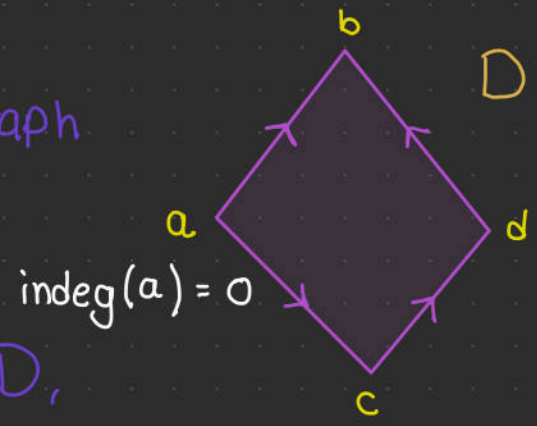
The indegree of a vertex v is the number of edges directed into v, and the outdegree is the # out.



$$\text{indeg}(v) + \text{outdeg}(v) = \deg(v)$$

Example 1.2.

Consider the graph



The edge (a, c) is in D, but (c, a) is not.

A network is a directed graph in which every edge e is assigned a non-negative integer $c(e)$.

The value $c(e)$ is the capacity of e and represents

the maximum amount of fluid / buses / packet etc. that can be transported along any edge.

In a **network flow** problem, the network has two distinguished vertices, the **source**, and the **sink**. We denote the source s, and the sink t. Note that $\text{indeg}(s) = 0$ and $\text{outdeg}(t) = 0$. All other vertices are **intermediate vertices**.

A **flow** in a network N (with source s and sink t) is a function $f : (V \times V) \to \mathbb{N}$ which satisfies the conditions:

i) For each e, $0 \leq f(e) \leq c(e)$ "the pipe cant burst"

ii) For each intermediate vertex v,

$$\sum_{e \in I(v)} f(e) = \sum_{e \in O(v)} f(e),$$

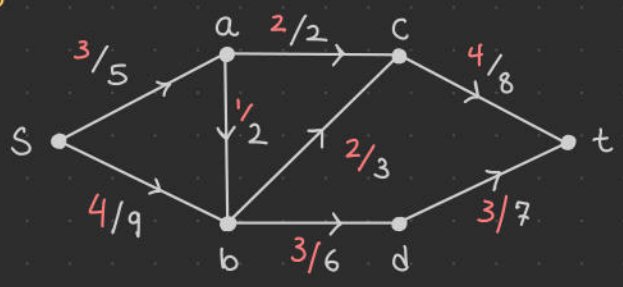
where $I(v)$ is the set of edges directed into v, and $O(v)$ directed out.

" we might have many edges going in and out, but no net loss or gain of material. "

The **value** of a flow is $\text{val}(f) = \sum$

$$\text{val}(f) = \sum_{e \in I(f)} = \sum_{e \in O(f)}$$

Example 1.3



" you can take the source or the sink and get back out what you origionally put in ".

Can you find a better flow?

The Maximum Flow Problem is to find a flow that minimises $\text{val}(f)$.

A first bound on the max flow is

$$\text{minimum} \begin{cases} \text{Val}(f) \leq \sum_{e \in O(s)} c(e) \\ \text{and} \\ \text{Val}(f) \leq \sum_{e \in I(t)} c(e) \end{cases}$$

Lecture 2

Flows & Cuts

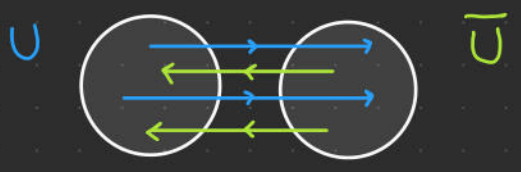
Let $N = (V, E)$ be a network, and let $U \subseteq V$ s.t $s \in U$ and $t \in V \setminus U$ (where $V \setminus U$ also known as $V - U$, is the set of vertices of V without the eles. of U). So $V = U$ union $\bar{U}$.

A cut in a flow network is a pair $(U, \bar{U})$ {with $\bar{U} = V \setminus U$}.

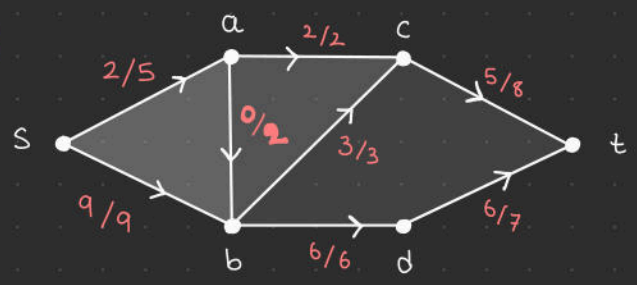
The capacity of the cut $(U, \bar{U})$ is defined as

$$c(U, \bar{U}) = \sum_{e \in A(U, \bar{U})} c(e),$$

where $A(U, \bar{U})$ denotes the set of edges directed from vertices in U to vertices in $\bar{U}$



Example 1.4



claimed max flow = 11

What are the capacities of the cuts

i) $U = \{s\}$ and $\bar{U} = \{a, b, c, d, t\}$

$c(U, \bar{U}) = c(sa) + c(sb) = 5 + 9 = 14$

ii) $U = \{s, b\}$, $\bar{U} = \{a, c, d, t\}$

$c(U, \bar{U}) = c(bc) + c(bd) + c(sa)$

$= 14$

iii) $U = \{s, c\}$, $\bar{U} = \{a, b, d, t\}$

$c(U, \bar{U}) = c(sa) + c(sb) + c(ct)$

$= 5 + 9 + 8$

$= 22$

iv) $U = \{s, a\}$, $\bar{U} = \{b, c, d, t\}$

$c(U, \bar{U}) = 13$

v) $U = \{s, a, b\}$

$c(U, \bar{U}) = 11$

Extending the def. of capacity of a cut $(U, \bar{U})$ for a flow f on network N, we have

$$f(U, \bar{U}) = \sum_{e \in A(U, \bar{U})} f(e),$$

is the flow of the cut.

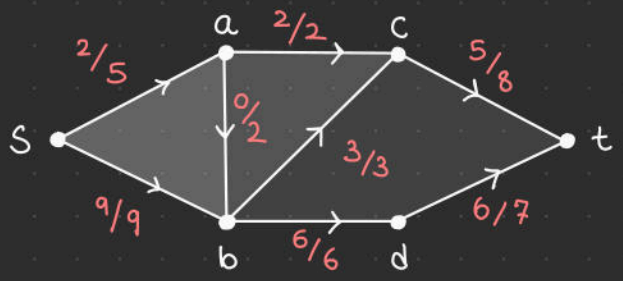
Q: how does this relate to the value of the flow?

Lemma 1.1

Let $\mathcal{N}$ be a flow network and let f be a flow on $\mathcal{N}$. If $(U, \bar{U})$ is a cut of $\mathcal{N}$, then

$$\text{val}(f) = f(U, \bar{U}) - f(\bar{U}, U) \qquad \text{(think: net flow)}$$

Example 1.5



Consider the following cut : $U = \{s, d\}$ and $\bar{U} = \{a, b, c, t\}$

Then $f(U, \bar{U}) = f(sa) + f(sb) + f(dt)$

$= 2 + 9 + 6$

$= 17,$

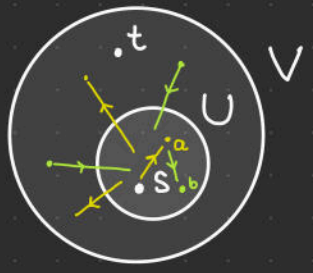
and $f(\bar{U}, U) = f(bd)$

$= 6$

$\Rightarrow \text{val}(f) = 17 - 6 = 11.$

(Proof of lemma 1.1)

Consider $f(U, V) - f(V, U)$ where V is the vertex set of $\mathcal{N}$.



Note that while (U, V) is not a cut, we have that $f(U, V) = \sum_{e \in A(U,V)} f(e)$.

particular vertex $u \in U$

Now $f(U, V) - f(V, U) = \sum_{u \in U} f(\{u\}, V) - \sum_{u \in U} f(V, \{u\})$

(inflow − outflow $= 0$) u $= \sum_{u \in U} \left(f(\{u\}, V) - f(V, \{u\}) \right)$

$$= f(\{s\}, V) - f(V, \{s\})$$
$$= \text{val}(f) - 0$$

Since for an intermediate vertex, flow in equals flow out and $f(V, \{s\}) = 0$ by definition, $\text{indeg}(s) = 0$ and $f(\{s\}, V) = \text{val}(f)$ by definition.

But $f(U, V) = f(U, U \text{ union } \bar{U})$

$= f(U, U) + f(U, \bar{U})$ since $U \cap \bar{U} = \phi$

and $f(V, U) = f(U \cup \bar{U}, U)$

$= f(U, U) + f(\bar{U}, U)$

Hence $\text{val}(f) = f(U, V) - f(V, U)$

$= f(U, U) + f(U, \bar{U}) - f(U, U) - f(\bar{U}, U)$

$= f(U, \bar{U}) - f(\bar{U}, U)$.

Corollary 1.2. Let N be a flow network, and let f be a flow on N. If $(U, \bar{U})$ is a cut, then $\text{val}(f) \leq c(U, \bar{U})$. {turns out to be equality}

"It doesnt matter which cut I take, the overall flow doesnt change"

Proof: Since $f(e) \leq c(e)$ for each edge of N, we have

$$f(U, \bar{U}) \leq c(U, \bar{U}),$$

and so, by Lemma 1.1,

$$\text{val}(f) = f(U, \bar{U}) - f(\bar{U}, U)$$
$$\leq f(U, \bar{U}) \quad (\text{as } f(\bar{U}, U) \geq 0)$$
$$\leq c(U, \bar{U})$$

In the previous example 1.5, found a flow f, s.t $val(f) = 11$. We found a cut $(U, \bar{U})$ (where $U = \{s, a, b\}$) where $c(U, \bar{U}) = 11$.

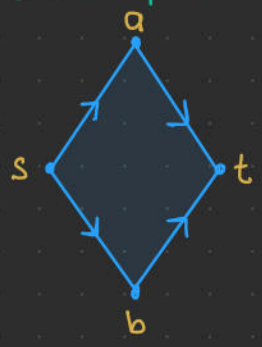
Flow Incrementing Paths

A path P is a graph $G = (V, E)$ that is an alternating sequence of vertices and edges:

$$v_1, e_1, v_2, e_2, \ldots, e_{k-1}, v_k,$$

such that e_i is incident to v_i and v_{i+1} (for $1 \leq i \leq k$) such that no vertex appears more than once.

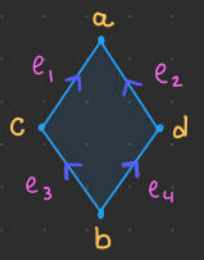
Example 1.7 (paths ignore arrow directions, but do have forward and reverse edges)



$P_1 = sat\,b$ — sa: forward edge

$P_2 = sb$

$P_3 = ta$ — reverse edge



$P = Ce_1, ae_2d$ — e_1: forward, e_2: backward

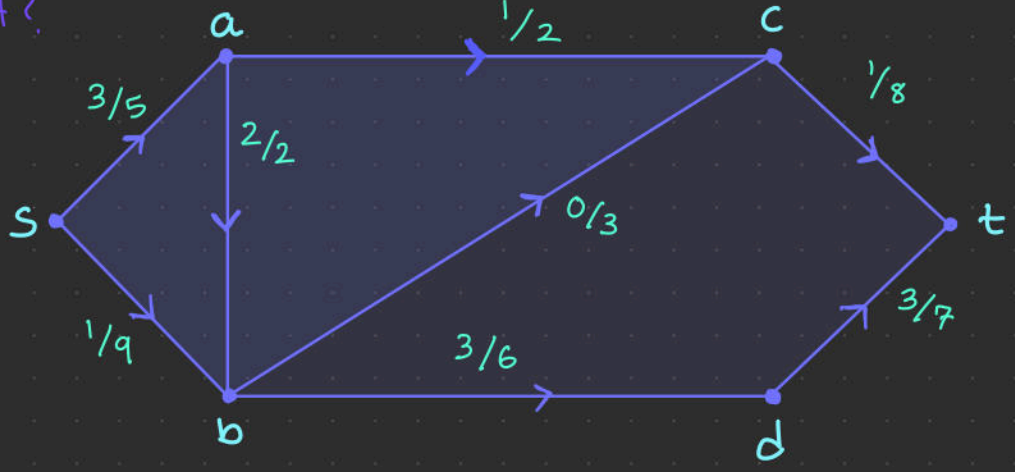
Let N be a network with capacity c. Let f be a flow on N and let P be a path from s to t. We say that P is a flow incrementing path if

i) $f(e) < c(e)$ for all forward edges of P (not equal to!)

ii) $f(e) > 0$ for all reverse edges of P.

i.e. can we increment?

Example 1.8



(i) P_1 = sact is a flow incrementing path

(ii) P_2 = sbact is a flow incrementing path

If P is a f-inc path, the increment is given by

$$i(P) = \min\{\{c(e) - f(e) : e \text{ is forward edge of } P\} \cup \{f(e) : e \text{ is a reverse edge}\}\},$$

where the min is taken over all edges e of P.

Example 1.9

Consider the network N from ex 1.8.

The increment of the path P = sbact is $\min\{9-1, 2, 2-1, 8-1\} = 1$.

Example 1.10:

By adding 1 to each of the forward edges in P and subtracting 1 from each of the reverse edges, we get a better flow.

$$Val(f) + 1 = 4 + 1 = 5$$

revisiting cor 1.2

$Val(f) \leq c(U, \bar{U})$

flow incrementing paths ↓ ? cuts

minimum diff. between

Prop 1.3. Let N be a flow network, and let f be the flow on N. Let P be a f-incrementing path of inc. i. Then by adding i to the flow of each forward edge and subtracting i from each reverse edge of P, we obtain a new flow with valve val(f) + i.

Theorem 1.4 (Max-flow, min-cut theorem).
The maximum value of a flow on a network N is equal to the minimum capacity of a cut.

Proof:
Let f be a maximum flow. Since f has max value, this means N contains no incrementing paths from s to t. However, starting at s we can construct a partial f-inc path (a path that satisfies the conditions for a f-inc path, but does not end at t). Let S be the set of vertices of N that can be reached by maximal partial f-inc paths (maximal means can't add more, maximum means biggest).

Note: We can have more than one path; we take their union. S is the set of vertices reached by maximal flows



Consider the cut $(S, \bar{S})$. We will show that $val(f) = c(s, \bar{S})$.

Let $e = (u, v)$ be an edge with $u \in S$, $v \in \bar{S}$. Then there is a partial f-inc. path from s to u.

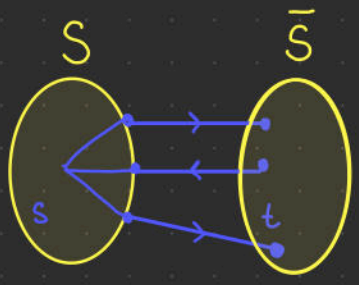
S is the set of end points for the flow inc paths, WTS that we can't go further.



If $f(e) < c(e)$, we could extend the path to v, contradicting that $v \in \bar{S}$. Hence $f(e) = c(e)$. {flowing forward}
Let $e' = (v, u)$ be an edge with $u \in S$ and $v \in \bar{S}$. If $f(e') > 0$, we could extend path from u to v, contradicting that

$v \in \bar{S}$. Hence $f(e') = 0$. {flowing backwards}

We've taken some path





cannot increment past here.

everything moving forward is at capacity, everything moving backward is zero.

By Lemma 1.1

$$\text{Val}(f) = f(S, \bar{S}) - f(S, \bar{S}).$$

But, from above,

$$f(S, \bar{S}) = c(S, \bar{S})$$

and $$f(S, \bar{S}) = 0.$$

That is, $\text{Val}(f) = c(S, \bar{S})$.

max (assumed) min

Since $\text{val}(f) \leq c(S, \bar{S})$ by cor 1.2, then $(S, \bar{S})$ is a minimum cut and the theorem follows.

The Ford-Fulkerson Algorithm

Based on the proof of the MFMC thm..

Output: Max flow in N

1. Begin with a flow on N
   (flow with $f(e)$, $\forall e \in E(N)$, is fine) {easy to start with 0 everywhere}
2. Starting at s, construct a tree by adding edges that are candidates for f-inc. paths.
   (called the f-inc tree) {if not backward edge then you can be in my tree}

* BFS / DFS (efficient algorithms ~ polynomial time)

③ If the tree reaches t, then the unique path in the tree joining s to t, is a f-incrementing path. Increment accordingly to make it a non-f-inc. path. GOTO 2.

If the tree does not reach t, then there is no f-inc. path. Therefore f has maximum value. END. Return f.

This algorithm is efficient: $O(|V||E|M)$, where M = max capacity of the network.

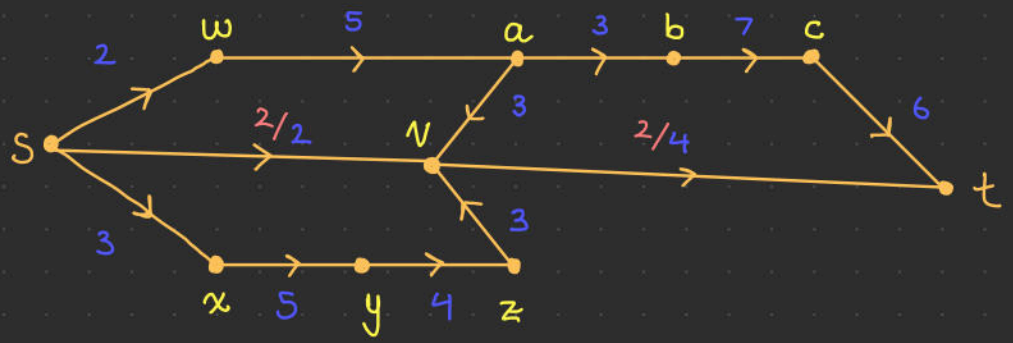
If tree: $|E| = n-1$, $|E| = \binom{n}{2} = O(n^2)$ so $O(n^3 m)$ worst case.

This method <u>always</u> finds the maximum flow.

What about the min cut?

The vertices of our final tree $V(T)$ are our minimal cut $(V(T), \overline{V(T)})$.

Home work exercise:



Start with flows of all zero, iterate through with mini pictures of each graph.

Flow incrementing path $P_1 = svt$ has $i(P_1) = 2$, so $val(f) = 0 + 2 = 2$.

Aside on proofs:

If ... then ... $P \to Q$

Direct proof: Assume P, prove Q.

Contrapositive: $\neg Q \to \neg P$

Contradiction: $(P \wedge \neg Q) \longrightarrow \mathbb{F}$

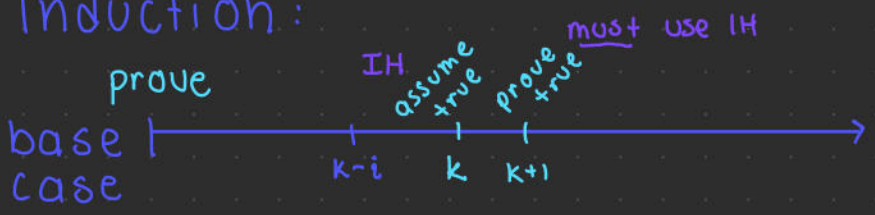
* Assume $(P \wedge \neg Q)$ and arrive at something false or contradict ourselves in some way.

Exhaustion: $(P \to Q) \iff (P_1 \to Q_1) \cup (P_2 \to Q_2) \cup \cdots \cup (P_k \to Q_k)$

for $P = P_1 \cup P_2 \cup \cdots \cup P_k$,

$Q = Q_1 \cup Q_2 \cup \cdots \cup Q_k$.

Induction:



heading off to infinity

1. prove base case
2. Take pair k, k+1
   * assume k true (IH)
   * prove k+1 using IH

☆ The use of IH to prove the next thing is the link between the dominos.

Strong: Assume base case $\longrightarrow$ k^{th} case is true

Weak: Assume k^{th} case is true

{ as soon as you need something less than k, strong

Tip: G has k+1 vertices:

* delete one vertex
* add it back and say why it still works

Homework:

Prove that a simple graph G has an even number of vertices of odd degree using all 5 proof techniques.